Universitas AMIKOM Yogyakarta

DATA SCIENCE PRESENTATION

Prediksi Ketinggian Ombak untuk Rekomendasi Waktu Kunjungan Aman di Kawasan Wisata Pantai

Data Science

Anggota Kelompok

MUHAMMAD ABIYYU RAHMAN (25.51.1738)

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AKMAL RAFI FADHILLAH (25.51.1810)

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Latar Belakang

- Ketinggian ombak merupakan data deret waktu yang berubah setiap jam dan memiliki pola non-linear yang sulit diprediksi secara manual.
- Pola ombak berubah secara dinamis mengikuti variasi temporal, sehingga diperlukan model prediksi yang mampu menangkap dependensi historis dan pola jangka pendek.
- Machine Learning dan Deep Learning mampu mempelajari pola ketinggian ombak dari data historis serta menghasilkan prediksi yang akurat.
- Perbandingan antar model diperlukan untuk menentukan pendekatan prediksi paling efektif yang dapat dimanfaatkan dalam konteks pengelolaan destinasi wisata pantai.

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Tujuan Penelitian

- Mengembangkan model prediksi ketinggian ombak.
- Membandingkan performa Machine Learning dan Deep Learning untuk menentukan model dengan akurasi prediksi terbaik.
- Menetapkan kategori keamanan ombak (Aman/Waspada/Bahaya) berdasarkan nilai prediksi ketinggian ombak.
- Menghasilkan rekomendasi waktu kunjungan aman pada pantai tertentu berdasarkan kategori keamanan ombak.



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Dataset Penelitian

- Dataset berisi ketinggian ombak per - tanggal, berbentuk wide format (Jam 1–24).
- Data ditransformasikan menjadi long format: Tanggal | Jam | Tinggi Ombak |
- Periode: harian (format time-series)
- Output yang dianalisis: Tinggi Ombak (meter)

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Paper 1

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Prediction of significant wave height using machine learning and its application to extreme wave analysis

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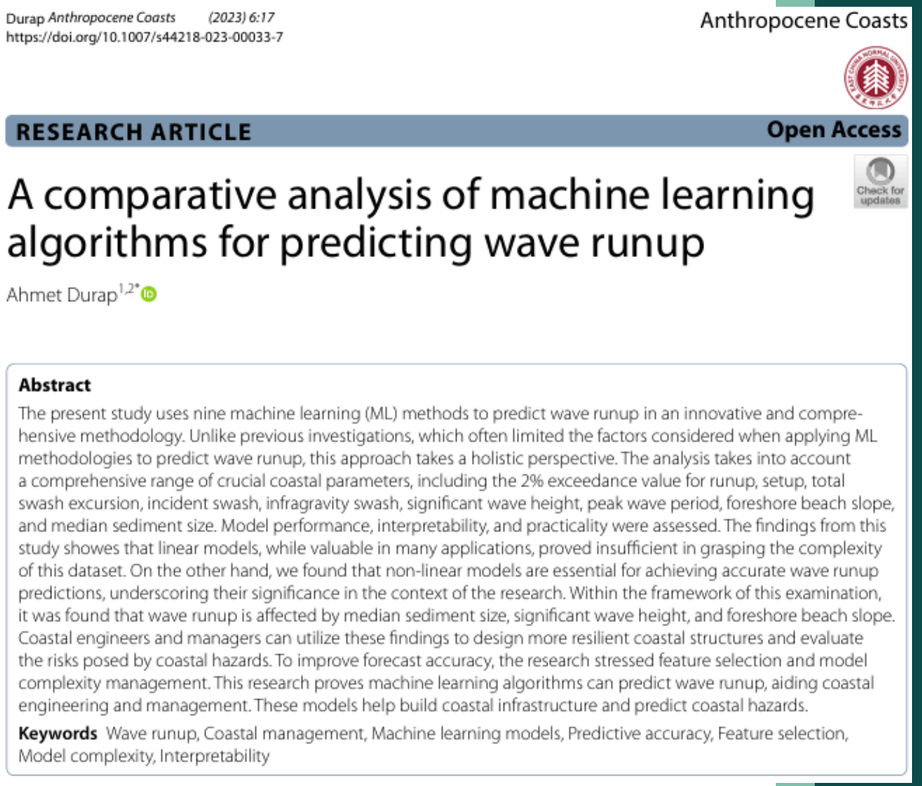
Waves of large size can damage offshore infrastructures and affect marine facilities. In coastal engineering studies, it is essential to have the probability estimates of the most extreme wave height expected during the lifetime of the structure. This study predicts significant wave height using the machine learning (ML) technique with generalized extreme value (GEV) theory and its application to extreme wave analysis. The wind speed, wind direction, sea temperature, and swell height data consisting of wave characteristics for 60 years has been obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF). While analyzing extreme waves, the block maxima approach in GEV was used to incorporate the seasonal variations present in the data. The estimated scale parameter, shape parameter, and location parameter of GEV are used in the ML model to predict the significant wave heights along with their return periods. The ML algorithms such as linear regression (LR), artificial neural networks (ANN), and support vector machines (SVM) are evaluated in terms of R^2 performance. The model comparison results suggested that the SVM model outperforms the LR and ANN models with an accuracy of 99.80%. Finally, the GEV analysis gives the extreme wave height results of 2.348, 3.470, and 4.713 m with a return period of 5, 20, and 100 yrs, respectively. Hence, the model developed is capable of predicting both significant wave height and extreme waves for the design of coastal structures.

Keywords. Significant wave height; generalized extreme value; artificial neural network; support vector machine; machine learning.

- Penelitian ini memprediksi significant wave height (SWH) menggunakan machine learning (LR, ANN, SVM).
- Input menggunakan data reanalysis ECMWF (WAM model): wind speed, wind direction, swell, wave parameters.
- SVM dengan Gaussian RBF kernel menjadi model terbaik dengan akurasi $R^2 = 0.998$ dan RMSE ≈ 3.95 cm.
- Selain prediksi gelombang normal, dilakukan extreme wave analysis menggunakan Generalized Extreme Value (GEV) untuk menghitung ketinggian ombak ekstrem.
- Fokus utama penelitian adalah desain struktur pantai, bukan area pariwisata atau rekomendasi keselamatan wisatawan.

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Paper 2



Durap *Anthropocene Coasts* (2023) 6:17
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Anthropocene Coasts

RESEARCH ARTICLE Open Access

A comparative analysis of machine learning algorithms for predicting wave runup

Ahmet Durap[1,2*]

Abstract

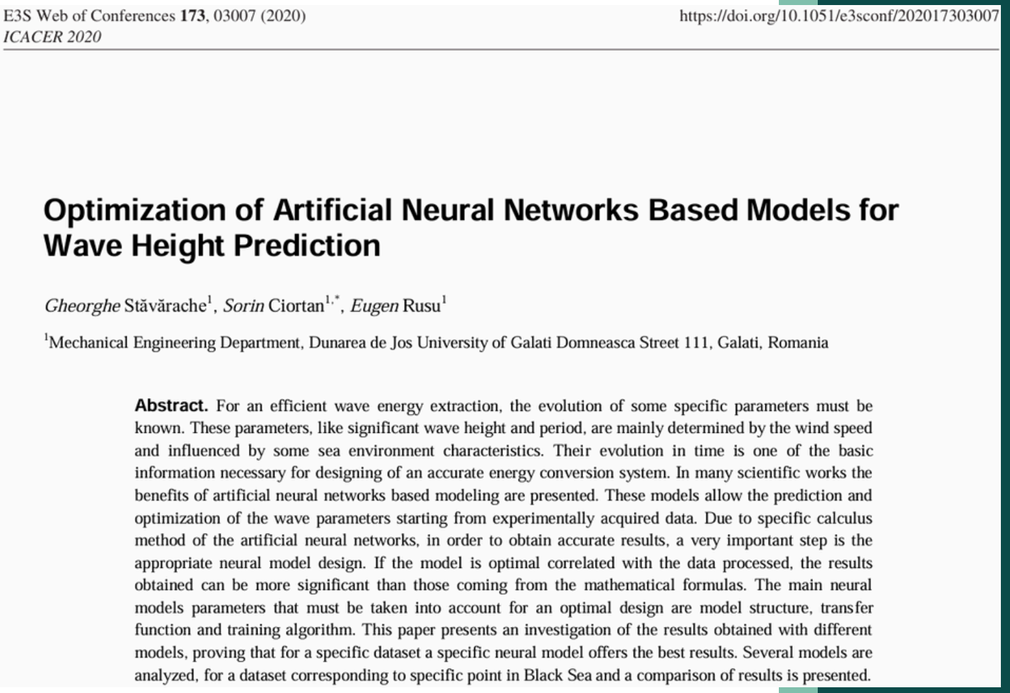
The present study uses nine machine learning (ML) methods to predict wave runup in an innovative and comprehensive methodology. Unlike previous investigations, which often limited the factors considered when applying ML methodologies to predict wave runup, this approach takes a holistic perspective. The analysis takes into account a comprehensive range of crucial coastal parameters, including the 2% exceedance value for runup, setup, total swash excursion, incident swash, infragravity swash, significant wave height, peak wave period, foreshore beach slope, and median sediment size. Model performance, interpretability, and practicality were assessed. The findings from this study showes that linear models, while valuable in many applications, proved insufficient in grasping the complexity of this dataset. On the other hand, we found that non-linear models are essential for achieving accurate wave runup predictions, underscoring their significance in the context of the research. Within the framework of this examination, it was found that wave runup is affected by median sediment size, significant wave height, and foreshore beach slope. Coastal engineers and managers can utilize these findings to design more resilient coastal structures and evaluate the risks posed by coastal hazards. To improve forecast accuracy, the research stressed feature selection and model complexity management. This research proves machine learning algorithms can predict wave runup, aiding coastal engineering and management. These models help build coastal infrastructure and predict coastal hazards.

Keywords Wave runup, Coastal management, Machine learning models, Predictive accuracy, Feature selection, Model complexity, Interpretability

- Penelitian ini membandingkan algoritma Machine Learning untuk prediksi wave runup.
- Dataset menggunakan 9 variabel kunci seperti significant wave height (Hs), swash components (Stt, Sinc, Sig), slope, setup, dll. (lihat Tabel 1 pada paper).
- Hasil menunjukkan model non-linear (Gradient Boosting, Stacking) memberikan performa dengan $R^2 \approx 0.999$.
- Analisis feature importance menunjukkan Hs, Stt, Sig sebagai faktor paling dominan terhadap runup.
- Penelitian menegaskan pentingnya ML untuk manajemen risiko pesisir, desain struktur pantai, dan mitigasi banjir.

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Paper 3



E3S Web of Conferences **173**, 03007 (2020) https://doi.org/10.1051/e3sconf/202017303007
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Optimization of Artificial Neural Networks Based Models for Wave Height Prediction

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Abstract. For an efficient wave energy extraction, the evolution of some specific parameters must be known. These parameters, like significant wave height and period, are mainly determined by the wind speed and influenced by some sea environment characteristics. Their evolution in time is one of the basic information necessary for designing of an accurate energy conversion system. In many scientific works the benefits of artificial neural networks based modeling are presented. These models allow the prediction and optimization of the wave parameters starting from experimentally acquired data. Due to specific calculus method of the artificial neural networks, in order to obtain accurate results, a very important step is the appropriate neural model design. If the model is optimal correlated with the data processed, the results obtained can be more significant than those coming from the mathematical formulas. The main neural models parameters that must be taken into account for an optimal design are model structure, transfer function and training algorithm. This paper presents an investigation of the results obtained with different models, proving that for a specific dataset a specific neural model offers the best results. Several models are analyzed, for a dataset corresponding to specific point in Black Sea and a comparison of results is presented.

- Penelitian ini membahas prediksi significant wave height (Hs) menggunakan berbagai Artificial Neural Network (ANN).
- Input data meliputi: month, water depth, wind speed.
- Model terbaik berdasarkan CC ≥ 0.95 dan RMSE ≤ 0.300 adalah ANN 5-5-9-8-1 + LM algorithm + sigmoid slope 1.
- Validasi menunjukkan performa: CC = 0.928 dan RMSE = 0.186 (lokasi P1), CC = 0.925 dan RMSE = 0.201.
- ANN juga digunakan untuk analisis importance: angin (wind speed) paling berpengaruh, disusul water depth.

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Research GAP

Gap 1: Minimnya komparasi lintas paradigma (deep learning vs machine learning).

Gap 2: Minimnya penelitian yang mengembangkan model prediksi ombak untuk tujuan keselamatan wisata atau rekomendasi jam kunjungan aman.

Gap 3: Minimnya model yang memproses data time-series secara murni untuk forecasting per-hari.



Exploratory Data Analysis (EDA)

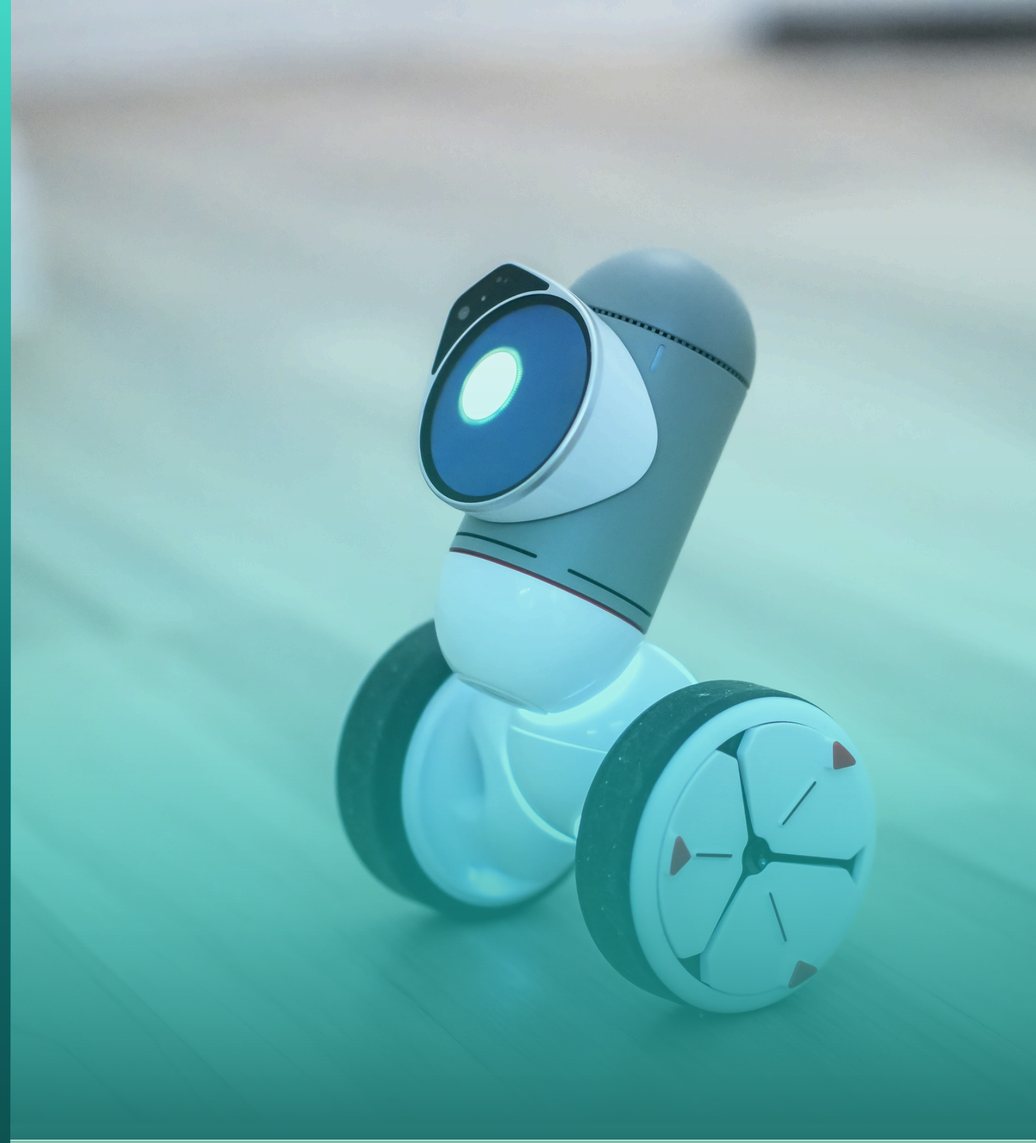
Analisis awal meliputi:

- Tren ketinggian ombak harian
- Pola musiman (harian/mingguan/bulanan)
- Heatmap intensitas ombak
- Distribusi ombak rendah vs tinggi
- Identifikasi jam rawan ombak tinggi

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Pemodelan

- Random Forest
- XGBoost
- LSTM



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Konversi Prediksi ke Kategori Keamanan

1. Prediksi ketinggian ombak ditransformasikan menjadi kategori risiko:

- < 1.0 m → Aman
- 1.0 – 2.0 m → Waspada
- > 2.0 m → Bahaya

2. Konversi ini menghasilkan:

- Jam aman kunjungan
- Jam yang harus dihindari wisatawan
- Jam yang butuh pengawasan ekstra
- Menjadi dasar rekomendasi keselamatan pantai



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Hasil Data Train

Random Forest

- MAE : 0.1791
- RMSE : 0.2587

XGBoost

- MAE : 0.1754
- RMSE : 0.2659

LSTM

- MAE : 0.2112
- RMSE : 0.2574

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Kesimpulan singkat

- Akurasi terbaik untuk data prediksi ketinggian ombak
- Lebih stabil
- Mudah dipakai untuk tanggal dimasa depan
- LEbih ringan dan cepat dibandingkan LSTM

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Kontribusi Penelitian

- Menghasilkan sistem prediksi ketinggian ombak yang dapat digunakan sebagai informasi keselamatan pantai.
- Menghasilkan model prediksi ketinggian ombak berbasis time-series yang dapat menjadi dasar penelitian lanjutan dalam bidang pemodelan maritim.
- Memberikan perbandingan komprehensif antara Machine Learning dan Deep Learning untuk prediksi time-series.
- Menyediakan rekomendasi jam kunjungan yang aman berdasarkan hasil prediksi ombak.
- Mendukung sistem keselamatan dan peringatan dini (early warning) berbasis data historis dan prediksi.

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TERIMA KASIH